

Chapter 9 — Big data and streaming

IoT and social platforms produce volume and velocity that traditional databases cannot absorb alone

Learning objectives

- By the end of this part

Defining big data and the Vs

- Big data refers to extremely large, complex datasets beyond conventional processing tools
- Classic dimensions include volume, the sheer size of the corpus

Example 9.23 — Petabyte-Scale Social and Sensor Volume

- Example 9.23 — hands-on module
- Example 9.23 illustrates volume
- At that size
- Explore the chapter example module
- View files: ``modules/chapter9/example23/``

Velocity, variety, veracity, and value

- Velocity rises with IoT, social
- Variety spans spreadsheets, JSON, XML, posts, video, and images
- Veracity is hard to assess at scale
- Value reminds teams that collecting volume without extractable insight wastes

Example 9.24 — Noise Filtering in Social Media Streams

- Example 9.24 — hands-on module
- Example 9.24 shows why social streams need filtering
- Collection pipelines therefore include quality gates, not only ingest
- Explore the chapter example module
- View files: ``modules/chapter9/example24/``

Streaming data versus batch processing

- Streaming data is generated continuously and must be processed as it arrives
- Streaming systems handle high-throughput flows in motion
- Stock feeds, IoT readings

Example 9.25 — Streaming Stock Sensor and Social Feeds

- Example 9.25 — hands-on module
- Example 9.25 lists common streaming sources
- Distributed systems split large datasets across machines so parallel processing can keep
- Explore the chapter example module
- View files: ``modules/chapter9/example25/``

Takeaways

- Big data is defined by volume, velocity, variety, veracity, and value, not size alone
- Social noise filtering protects insight quality
- Social collection at scale

Next

- Complete the quiz for this part
- The next part surveys collection challenges and tooling